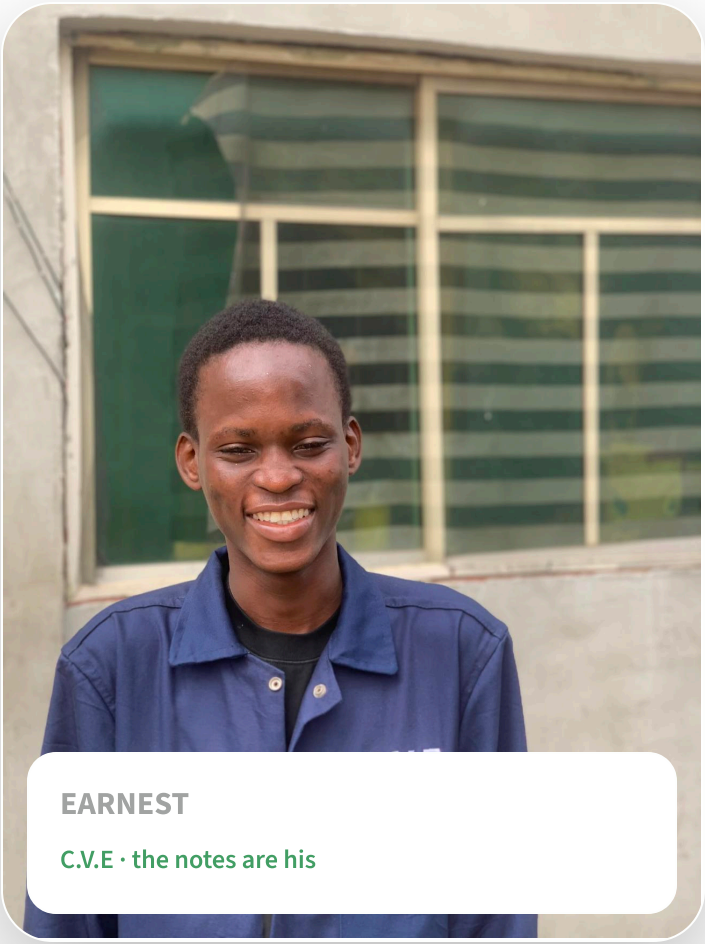
ENGINEERING · FLUID MECHANICS

Understand it. *Don't just memorize it.*

A student-first guide by EARNEST. All 21 chapters of the degree course, from the properties of a fluid to presses, rams and couplings. The idea first, then the formula.

Start with chapter 01 Test yourself



21

Chapters, the full course

18

Formulas with the assumption written beside them

20

Exam-style questions with the arithmetic shown

6

Calculators that show the substitution, not just the answer

HOW TO STUDY

Four passes. Then the paper.

Use the same order on every chapter. Skipping to the formula is how the formula gets forgotten.

01

Read the idea

One paragraph first. If you cannot say the chapter without a formula, you do not have it yet.

02

Walk the topic list

Every heading you will meet in class is there in plain language. Tick them. Do not skip a heading because the formula looks familiar.

03

Do the number

04

Then the quiz

Substitute before you rearrange. The worked example uses g = 9.81 and shows the unit of the answer.

Close the notes. The explanations show the arithmetic, so a wrong pick still teaches the step.

SYLLABUS

Chapters

Open a chapter for the idea, every topic in that chapter, one worked number, and the exam trap.

All | At rest | In motion | Machines

Search pressure, Bernoulli, pump, Manning…

Showing 21 of 21

0 marked studied

Chapter 01 · At rest

Properties of fluids

The few numbers the rest of the course is built from: density, viscosity, compressibility and surface tension.

Open chapter

Chapter 02 · At rest

Pressure and its measurement

How pressure grows with depth, and how manometers turn that into a reading.

Open chapter

Chapter 03 · At rest

Hydrostatic forces on surfaces

The total push on a wall, gate or curve, and the point where that push acts.

Open chapter

Chapter 04 · At rest

Buoyancy and floatation

Why a body floats, and why a small tilt sometimes rights it and sometimes capsizes it.

Open chapter

Chapter 05 · In motion

Kinematics and ideal flow

How to describe motion, then the imaginary frictionless patterns: sources, sinks, vortices and flow past a cylinder.

Open chapter

Chapter 06 · In motion

Dynamics of fluid flow

Euler, Bernoulli, venturi, orifice meter, Pitot tube, and the momentum equation.

Open chapter

Chapter 07 · In motion

Orifices and mouthpieces

Holes in tanks: how much contracts, how fast it leaves, and how long the tank takes to empty.

Open chapter

Chapter 08 · In motion

Notches and weirs

Openings in the side of a channel that measure discharge from the head over the crest.

Open chapter

Chapter 09 · In motion

Viscous flow

Slow, layered motion: pipes, gaps between plates, bearings and a sphere falling through oil.

Open chapter

Chapter 10 · In motion

Turbulent flow

Reynolds' dye experiment, turbulent shear, and why a smooth pipe and a rough pipe behave differently.

Open chapter

Chapter 11 · In motion

Flow through pipes

Major losses, minor losses, siphons, branches, nozzles, water hammer and a pipe network.

Open chapter

Chapter 12 · In motion

Dimensional and model analysis

How to shrink a river, a ship or a valve into a lab test that still means something.

Open chapter

Chapter 13 · In motion

Boundary layer flow

The thin slowed skin of fluid on a wall, and what happens when it lets go.

Open chapter

Chapter 14 · In motion

Forces on submerged bodies

Drag, lift, terminal velocity, and why a spinning ball curves.

Open chapter

Chapter 15 · In motion

Compressible flow

When density changes: the speed of sound, Mach number, and nozzles that must diverge to go faster.

Open chapter

Chapter 16 · In motion

Flow in open channels

Rivers, drains and flumes: Chezy, Manning, the best cross-section, specific energy and the hydraulic jump.

Open chapter

Chapter 17 · Machines

Impact of jets

The force when a jet hits a plate or a vane, fixed or moving, and jet propulsion.

Open chapter

Chapter 18 · Machines

Turbines

Pelton, Francis and Kaplan: the wheel that fits the head and the flow, plus draft tubes and specific speed.

Open chapter

Chapter 19 · Machines

Centrifugal pumps

Impellers that throw water outward: head, efficiency, priming, specific speed and NPSH.

Open chapter

Chapter 20 · Machines

Reciprocating pumps

A piston that traps a volume: slip, the indicator diagram, acceleration head and air vessels.

Open chapter

Chapter 21 · Machines

Fluid systems

Presses, accumulators, intensifiers, rams, lifts, cranes, couplings, torque converters and the air-lift pump.

Open chapter

LEARN

Six ideas that carry the course

If these six sit still in your head, the later formulas are variations, not new subjects.

What a fluid is

Pressure

Bernoulli

Reynolds

Machines

Free surfaces

WHAT A FLUID IS

It moves if you shear it

> A solid leans and stops. A fluid keeps sliding for as long as the shear is there. Liquids

> and gases both do this. That is the whole subject, before any machine is added.

Viscosity, pipe friction, drag and lubrication are all this one fact with a different geometry.

$$\tau = \mu \, du/dy$$

τ — shear stress, Pa

μ — dynamic viscosity, Pa·s

du/dy — velocity gradient, 1/s

HOLD ONTO THIS

Water at 20°C needs only about 0.2 Pa to sustain a gradient of 200 s^{-1}. Honey needs hundreds of times more.

REFERENCE

Formulas, with the catch

A formula without its assumption is how a correct-looking answer loses the mark.

Newtonian shear

$$\tau = \mu \, du/dy$$

Dynamic viscosity. Kinematic viscosity is ν = μ/ρ.

Droplet and bubble

$$\Delta p = 4\sigma/d \text{ or } 8\sigma/d$$

4σ/d for a liquid droplet, 8σ/d for a soap bubble, 2σ/d for a jet. d is diameter.

Hydrostatic pressure

$$p = \rho \, g \, h$$

Force on a plane surface

$$F = \rho \, g \, h_c \, A$$

Gauge pressure at depth h. Absolute pressure adds the atmosphere.

h_c is the centroid depth. The force acts at the centre of pressure, lower down.

Buoyancy and BM

$$F_B = \rho g V \quad BM = I / V$$

GM = BM − BG when G is above B. Positive GM is stable for a small heel.

Continuity

$$Q = A_1 V_1 = A_2 V_2$$

Constant density. Halving the diameter multiplies the speed by four.

Bernoulli head

$$p/(\rho g) + V^2/(2g) + z = H$$

Steady, inviscid, incompressible, one streamline, no machine and no loss.

Orifice

$$Q = Cd\ a \sqrt{(2gH)}$$

Cd = Cc × Cv. A sharp small orifice has Cd near 0.62.

Rectangular weir

$$Q = (2/3)\ Cd\ L \sqrt{(2g)}\ H^{(3/2)}$$

Triangular: (8/15) Cd tan(θ/2) √(2g) H^(5/2).

Pipe friction, two f’s

$$hf = f\ L\ V^2 / (2gD)$$

This f is Darcy, 64/Re when laminar. Notes that write 4f use an f four times smaller.

Water hammer

$$\Delta p = \rho\ V\ c$$

Sudden closure. Slow closure, time t longer than 2L/c: Δp ≈ ρ V L / t.

Manning and Chezy

$$V = (1/n)\ R^{(2/3)}\ S^{(1/2)}$$

SI Manning. Chezy is V = C √(R S). R = A/P.

Hydraulic jump

Drag

$$y_2/y_1 = \tfrac{1}{2}\left(-1 + \sqrt{1 + 8\,Fr_1^2}\right)$$

Rectangular channel. Fr_1 must be greater than 1 or there is no jump.

$$F_D = \tfrac{1}{2} C_D \rho A V^2$$

Stokes sphere, Re ≪ 1: F_D = 3 π μ V d.

Sound speed

$$c = \sqrt{\gamma R T}$$

Perfect gas, adiabatic wave. Mach number M = V/c.

Jet on a flat plate

$$F = \rho A V^2$$

Fixed plate. One moving plate: $\rho A(V-u)^2$. A series of vanes: $\rho AV(V-u)$.

Fluid power

$$P = \rho g Q H$$

Divide by pump efficiency for shaft input. Multiply by turbine efficiency for shaft output.

Reciprocating discharge

$$Q = A L N / 60$$

Single-acting theoretical flow. Double-acting is twice this, before slip.

Symbol sheet

YOUR PROGRESS

Kept on this device

0 of 21 chapters marked. Nothing is uploaded.

Best quiz 0/20

Reset progress

QUIZ

Question 1 of 20 · best 0/20

Which statement defines a fluid?

- It has zero viscosity
- Any shear stress makes it deform continuously
- It must be a liquid, not a gas
- Its density never changes

CALCULATOR

$g = 9.81\ m/s^2$

Substitute, then read the number

Pressure | Discharge | Reynolds | Bernoulli | Pipe loss | Buoyancy

Density ρ (kg/m³)	Depth h (m)
1000	2

19.62 kPa

Gauge pressure under a free surface open to the atmosphere.

$$p = \rho g h$$

$$p = 1{,}000 \times 9.81 \times 2$$

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p = 19,620 Pa = 19.62 kPa
Same pressure as 2 m of fresh water.
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These check the method. Match your course’s rounding, and remember a Darcy f is four times a Fanning f. Sample check: water jet 0.002 m^2 at 10 m/s gives 200 N on a flat plate.

REFERENCE

The guide is by EARNEST

C.V.E. These notes follow the full degree course, in his own words, so a chapter can be read before the lecture and still make sense on the bus home.



EARNEST



C.V.E